MATHEMATICS PART 2 (GEOMETRY) | CLASS 10 | MAHARASHTRA SSC BOARD

# Chapter 7: Mensuration

**MODEL ANSWER SHEET / ANSWER KEY**

| 46 | 8 | 5 |
|---|---|---|
| QUESTIONS | IN TEXTBOOK | FIGURES |

**HOW TO USE**
*Cover the answer box ✓ and answer first. A* **TEXTBOOK** *strip means the same question is in your Balbharati Math-2 textbook, at the set and page shown. Tick ☐ each section once revised.*

| Q.1 (A) | Q.1 (B) | Q.2 (A) | Q.2 (B) | Q.3 (A) | Q.3 (B) | Q.4 | Q.5 |
|---|---|---|---|---|---|---|---|
| Multiple Choice | One-step Solve | Activities | Short Solve | Activities | Solve / Prove | Long Solve / Prove | Creative / HOTS |
| 10 Q | 5 Q | 5 Q | 6 Q | 6 Q | 5 Q | 4 Q | 5 Q |

## Q.1 (A) Multiple Choice — 1 Mark each

10 QUESTIONS · 2 ☐ Revised

**Q1** If the dimensions of a cuboid in cm are $16 \times 14 \times 20$, then its total surface area is _____.

(a) $4480\ \text{cm}^2$ (b) $1648\ \text{cm}^2$
(c) $824\ \text{cm}^2$ (d) $1740\ \text{cm}^2$

**✓ ANSWER / SOLUTION**

$1648\ \text{cm}^2$

**Explanation:**

Given: $l = 16$ cm, $b = 14$ cm, $h = 20$ cm

Total Surface Area of Cuboid $= 2(lb + bh + hl)$

$= 2(16 \times 14 + 14 \times 20 + 20 \times 16)$

$= 2(224 + 280 + 320) = 2(824) = 1648\ \text{cm}^2$

**Q2** The total surface area of hemisphere is $300\pi\ \text{cm}^2$, then find its radius.

(a) 8 cm (b) 12 cm
(c) 10 cm (d) 9 cm

**✓ ANSWER / SOLUTION**

10 cm

**Explanation:**

Total surface area of hemisphere $= 3\pi r^2$

$3\pi r^2 = 300\pi \implies 3r^2 = 300 \implies r^2 = 100 \implies r = 10$ cm

**Q3** Find the perimeter of a sector of a circle if its measure is $90^\circ$ and radius is $7$ cm.

TEXTBOOK **Problem Set 7 · Q.1 (3) · Page 160**

CORRECTED $25$ cm → $36$ cm — option (c) repeated option (a); Balbharati Problem Set 7 Q.1 (3) gives 36 cm

(a) $25$ cm (b) $44$ cm
(c) $36$ cm (d) $56$ cm

**ANSWER / SOLUTION**

$25$ cm

**Explanation:**

Length of arc $l = \frac{\theta}{360^\circ} \times 2\pi r = \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 7 = \frac{1}{4} \times 44 = 11$ cm

Perimeter of sector $= l + 2r = 11 + 2(7) = 11 + 14 = 25$ cm

**Q4** The radius of a cone is $7$ cm and height is $24$ cm. What is its curved surface area?

TEXTBOOK **Problem Set 7 · Q.1 (4) · Page 160**

(a) $550$ cm$^2$ (b) $110$ cm$^2$
(c) $440$ cm$^2$ (d) $330$ cm$^2$

**ANSWER / SOLUTION**

$550$ cm$^2$

**Explanation:**

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$ cm

Curved Surface Area $= \pi r l = \frac{22}{7} \times 7 \times 25 = 550$ cm$^2$

**Q5** For a cuboid $l^2 + b^2 + h^2 = 484$ cm$^2$ then what is the length of its diagonal?

(a) $12$ cm (b) $22$ cm
(c) $11$ cm (d) $24$ cm

**ANSWER / SOLUTION**

$22$ cm

**Explanation:**

Length of diagonal of a cuboid $= \sqrt{l^2 + b^2 + h^2} = \sqrt{484} = 22$ cm

**Q6** If the radius of the sector is $6$ and length of its corresponding arc is $14$, then area of the sector is _____.

(a) $35$ (b) $84$ (c) $42$ (d) $24$

**ANSWER / SOLUTION**

$42$

**Explanation:**

Area of sector $= \frac{l \times r}{2} = \frac{14 \times 6}{2} = 42$

**Q7** Find the ratio of the volumes of a cylinder and cone having equal radius and equal height.

(a) $1:3$ (b) $3:1$ (c) $2:1$ (d) $1:2$

**ANSWER / SOLUTION**

$3:1$

**Explanation:**

Volume of Cylinder $= \pi r^2 h$; Volume of Cone $= \frac{1}{3}\pi r^2 h$

Ratio $= \frac{\pi r^2 h}{\frac{1}{3}\pi r^2 h} = \frac{3}{1} = 3:1$

**Q8** The height of cone is $12$ cm and radius is $5$ cm then its slant height is _____.

(a) $17$ cm (b) $60$ cm
(c) $7$ cm (d) $13$ cm

**ANSWER / SOLUTION**

$13$ cm

**Explanation:**

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm

**Q9** What is volume of the bath tub in litres if its volume in $\text{cm}^3$ is $2058$?

(a) $2058$ (b) $20.58$ (c) $2.058$ (d) $205.8$

**ANSWER / SOLUTION**

$2.058$

**Explanation:**

$1$ litre $= 1000\ \text{cm}^3 \implies$ Volume in litres $= \frac{2058}{1000} = 2.058$ litres

**Q10** A solid sphere of diameter $6$ cm is melted and drawn into a wire of radius $4$ mm. The length of wire is _____.

(a) $90\pi$ cm (b) $90$ cm
(c) $900$ m (d) $225$ cm

**ANSWER / SOLUTION**

$225$ cm

**Explanation:**

Sphere radius $R = 3$ cm. Wire radius $r = 4$ mm $= 0.4$ cm.

Volume of wire (cylinder) $=$ Volume of sphere

$\pi r^2 h = \frac{4}{3}\pi R^3 \implies (0.4)^2 h = \frac{4}{3}(3)^3 = 36 \implies 0.16h = 36 \implies h = \frac{36}{0.16} = 225$ cm

## Q.1 (B) One-step Solve

1 Mark each

5 QUESTIONS · 1 Revised

**Q1** Find the area of the segment of a circle of radius $7$ cm whose corresponding sector has a central angle of $60^\circ$ ( $\pi = 3.14$).

**ANSWER / SOLUTION**

Radius $r = 7$ cm, central angle $\theta = 60^\circ$.

Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 7^2 = \frac{1}{6} \times 154 = 25.67\ \text{cm}^2$

Area of $\Delta = \frac{\sqrt{3}}{4} r^2 = \frac{1.732}{4} \times 49 = 21.22\ \text{cm}^2$

Area of segment $=$ Area of sector $-$ Area of $\Delta = 25.67 - 21.22 = 4.45\ \text{cm}^2$

**Q2** Area of a sector of a circle of radius $15$ cm is $30\ \text{cm}^2$. Find the length of the arc of the sector.

TEXTBOOK Practice Set 7.3 · Q.5 · Page 154

**ANSWER / SOLUTION**

Area of sector $= \frac{l \times r}{2}$

$30 = \frac{l \times 15}{2} \implies 15l = 60 \implies l = 4$ cm

$\therefore$ Length of the arc of the sector is $4$ cm.

**Q3** Radius of sector of a circle is $5\ \text{cm}$ and length of its arc is $2.8\ \text{cm}$. Find the area of the sector.

**ANSWER / SOLUTION**

Area of sector $= \frac{l \times r}{2} = \frac{2.8 \times 5}{2} = 1.4 \times 5 = 7\ \text{cm}^2$

**Q4** Find the surface area of sphere of radius $4.2\ \text{cm}$.

**ANSWER / SOLUTION**

Surface area of sphere $= 4\pi r^2 = 4 \times \frac{22}{7} \times 4.2 \times 4.2 = 4 \times 22 \times 0.6 \times 4.2 = 221.76\ \text{cm}^2$

**Q5** The radii of circular ends of frustum of a cone are $20\ \text{cm}$ and $12\ \text{cm}$ and its height is $6\ \text{cm}$. Find the slant height of frustum.

**ANSWER / SOLUTION**

$r_1 = 20\ \text{cm}$, $r_2 = 12\ \text{cm}$, $h = 6\ \text{cm}$

Slant height $l = \sqrt{h^2 + (r_1 - r_2)^2} = \sqrt{6^2 + (20-12)^2} = \sqrt{36 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10\ \text{cm}$

## Q.2 (A) Activities — 2 Marks each

5 QUESTIONS · 1 — Revised

**Q1** If the heights of two cylinders are equal and their radii are in the ratio of $7 : 5$ then complete the activity to find the ratio of their volumes.

**ANSWER / SOLUTION**

Volume of cylinder $= \boxed{\pi r^2 h}$ .....(Formula)

Heights are equal: $h_1 = h_2$

$\frac{r_1}{r_2} = \frac{7}{\boxed{5}}$

$\frac{V_1}{V_2} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{7}{5}\right)^2 = \boxed{\frac{49}{25}}$

**Q2** A road roller of length $3l\ \text{m}$ and radius $\frac{l}{3}\ \text{m}$ can cover field in 100 revolutions moving once over. Complete the activity to find the area of the field in terms of $l$.

**ANSWER / SOLUTION**

Curved surface area $= 2\pi r h$

$h = \boxed{3l}\ \text{m}$, $r = \boxed{\frac{l}{3}}\ \text{m}$

$\therefore$ Curved surface area of cylinder $= 2\pi \times \frac{l}{3} \times 3l = \boxed{2\pi l^2}$

$\therefore$ Area of field $= 2\pi l^2 \times 100 = \boxed{200\pi l^2}\ \text{m}^2$

**Q3** Find the volume of greatest right circular cone, which can be cut from a cube of a side $4\ \text{cm}$.

**ANSWER / SOLUTION**

Let, diameter of cone be edge of the $\boxed{\text{cube}}$.

$\therefore$ Side $s = \boxed{4}\ \text{cm}$

$\therefore h = \boxed{4}\ \text{cm}$, $r = 2\ \text{cm}$

Volume of cone $= \boxed{\frac{1}{3}\pi r^2 h}$ .....(Formula)

$V = \frac{1}{3}\pi(\boxed{2})^2 \times 4$

$V = \boxed{\frac{16\pi}{3}}\ \text{cm}^3$

**Q4** The measure of central angle of circle is $150^\circ$ and the radius of circle is $21$ cm. To find the length of the arc complete the following activity.

TEXTBOOK **Solved Example** · Ex. (1) (length of arc) · **Page 152**

**ANSWER / SOLUTION**

Length of arc $l = \frac{\theta}{360^\circ} \times \boxed{2\pi r}$

$= \frac{150^\circ}{\boxed{360^\circ}} \times \boxed{21} \times \frac{22}{7} \times 2$

$= \boxed{55}$ cm

**Q5** A right circular cone is such that the angle at its vertex is $90^\circ$ and its base radius is $49$ cm. Complete the activity to find the curved surface area of the cone.

**ANSWER / SOLUTION**

Let slant height be $x$ and $\perp$ will bisect vertex angle.

$\therefore \Delta ABC$ is right angle $\Delta$

$\therefore x^2 + x^2 = (49\sqrt{2})^2 \implies 2x^2 = 4802 \times 2 \implies x = 49\sqrt{2}$

Curved surface area of cone $= \boxed{\pi rl}$ ...(Formula)

$= \frac{22}{7} \times 49 \times \boxed{49\sqrt{2}}$

$= 22 \times 7 \times 49\sqrt{2} = \boxed{7546\sqrt{2}}$ cm$^2$

## Q.2 (B) Short Solve

2 Marks each

6 QUESTIONS · 1 ☐ Revised

**Q1** The diameter of a garden roller is $1.4$ m and it is $2$ m long. How much area will it cover in 5 revolutions? ($\pi = \frac{22}{7}$)

**ANSWER / SOLUTION**

Radius $r = \frac{1.4}{2} = 0.7$ m, height $h = 2$ m.

Area covered in 1 revolution $=$ Curved surface area $= 2\pi rh = 2 \times \frac{22}{7} \times 0.7 \times 2 = 8.8$ m$^2$

Area covered in 5 revolutions $= 5 \times 8.8 = 44$ m$^2$

**Q2** Find the surface area of a sphere of radius $3.5$ cm.

**ANSWER / SOLUTION**

Surface area of sphere $= 4\pi r^2 = 4 \times \frac{22}{7} \times 3.5 \times 3.5 = 4 \times 22 \times 0.5 \times 3.5 = 154$ cm$^2$

**Q3** The volume of a cube is $1000$ cm$^3$. Find its total surface area.

**ANSWER / SOLUTION**

Volume of cube $= l^3 = 1000 \implies l = 10$ cm

Total surface area $= 6l^2 = 6 \times (10)^2 = 6 \times 100 = 600$ cm$^2$

**Q4** A cone of height $24$ cm has a plane base of surface area $154$ cm$^2$. Find its volume.

**ANSWER / SOLUTION**

Base area $\pi r^2 = 154$ cm$^2$, height $h = 24$ cm.

Volume of cone $= \frac{1}{3} \times$ Base area $\times h = \frac{1}{3} \times 154 \times 24 = 154 \times 8 = 1232$ cm$^3$

**Q5** Find the surface area of hemisphere with radius $10$ cm ($\pi = 3.14$).

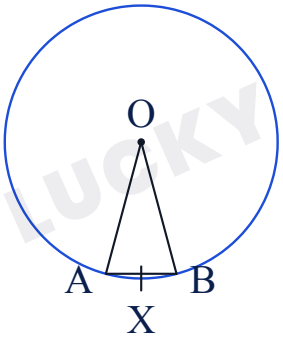
**ANSWER / SOLUTION**

Curved surface area $= 2\pi r^2 = 2 \times 3.14 \times (10)^2 = 6.28 \times 100 = 628$ cm$^2$

Total surface area $= 3\pi r^2 = 3 \times 3.14 \times (10)^2 = 9.42 \times 100 = 942$ cm$^2$

**Q6** In figure, point O is the center of the circle. $\angle AOB = 30°$, $OA = 12$ cm. Find the area of segment AXB ($\pi = 3.14$).

TEXTBOOK **Solved Example · Ex. (1) · Page 157**




**ANSWER / SOLUTION**

$r = 12$ cm, $\theta = 30°$.

Area of sector O-AXB $= \frac{\theta}{360°} \times \pi r^2 = \frac{30°}{360°} \times 3.14 \times 12^2 = \frac{1}{12} \times 3.14 \times 144 = 37.68$ cm$^2$

Area of $\Delta AOB = \frac{1}{2} r^2 \sin\theta = \frac{1}{2} \times 144 \times \sin 30° = 72 \times \frac{1}{2} = 36$ cm$^2$

Area of segment AXB = Area of sector − Area of $\Delta = 37.68 - 36 = 1.68$ cm$^2$

## Q.3 (A) Activities

3 Marks each

6 QUESTIONS · 2 ☐ Revised

**Q1** The circumferences of circular faces of frustum are $132$ cm and $88$ cm and its height is $24$ cm, to find the curved surface area of the frustum complete the following activity ($\pi = \frac{22}{7}$).

TEXTBOOK **Practice Set 7.2 · Q.3 · Page 148**

**ANSWER / SOLUTION**

Circumference$_1 = 2\pi r_1 = 132 \implies r_1 = \frac{132}{2\pi} = \boxed{21 \text{ cm}}$

Circumference$_2 = 2\pi r_2 = 88 \implies r_2 = \frac{88}{2\pi} = \boxed{14 \text{ cm}}$

Slant height of frustum $(l) = \sqrt{h^2 + (r_1 - r_2)^2} = \sqrt{\boxed{24}^2 + \boxed{7}^2} = \sqrt{576 + 49} = \boxed{25}$ cm

$\therefore$ Curved surface area of frustum $= \pi(r_1 + r_2)l = \frac{22}{7}(21 + 14) \times 25 = \boxed{2750}$ cm$^2$

**Q2** The circumferences of circular faces of a frustum are $132$ cm and $88$ cm and its slant height is $25$ cm. Complete the activity to find the total surface area of frustum.

**ANSWER / SOLUTION**

$2\pi r_1 = 132 \implies r_1 = \boxed{21}$ cm, $\quad 2\pi r_2 = 88 \implies r_2 = \boxed{14}$ cm

CSA of frustum $= \pi(r_1 + r_2)l = \frac{22}{7}(21 + 14) \times 25 = \boxed{2750}$ cm$^2$

TSA of frustum $=$ CSA $+ \pi r_1^2 + \pi r_2^2 = 2750 + \frac{22}{7}(21^2) + \frac{22}{7}(14^2) = 2750 + \boxed{1386} + \boxed{616} = \boxed{4752}$ cm$^2$

**Q3** The area of a minor sector of a circle is $3.85$ cm$^2$. The measure of its central angle is $36°$. Find the radius of circle by completing the activity.

TEXTBOOK **Practice Set 7.3 · Q.11 · Page 155**

**ANSWER / SOLUTION**

Area of minor sector $= 3.85$ cm$^2$, measure of central angle $(\theta) = 36°$.

Area of minor sector $= \frac{\theta}{360°} \times \boxed{\pi r^2}$

$3.85 = \frac{36°}{360°} \times \frac{22}{7} \times r^2 \implies r^2 = \frac{\boxed{3.85} \times \boxed{360°} \times \boxed{7}}{\boxed{36° \times 22}} = 12.25$

$r = \boxed{3.5 \text{ cm}}$

**Q4** **A hollow hemisphere bowl of thickness $1$ cm has an inner radius of $6$ cm. Find the volume of metal required to make the bowl.**

**ANSWER / SOLUTION**

Inner radius $r = 6$ cm, thickness $t = 1$ cm $\implies$ Outer radius $R = 6 + 1 = 7$ cm.

Volume of metal required $= \frac{2}{3}\pi R^3 - \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times \boxed{7}^3 - \frac{2}{3} \times \frac{22}{7} \times \boxed{6}^3$

$= \frac{44}{21}(\boxed{343} - 216) = \frac{44}{21} \times (\boxed{343} - \boxed{216}) = \frac{44}{21} \times \boxed{127} = \frac{5588}{21}$ cm$^3$

**Q5** **In the figure given below, ABCD is a square of side $7$ cm. BD is an arc of a circle of radius AB. What is the area of shaded region?**

**ANSWER / SOLUTION**

Area of shaded region $= 2(\text{Area of sector BAD} - \text{Area of } \Delta ABD)$

$= 2\left[\frac{90^\circ}{360^\circ} \times \boxed{\frac{22}{7}} \times 7^2 - \boxed{\frac{1}{2}} \times 7^2\right] = 2\left[\frac{1}{4} \times 154 - \boxed{24.5}\right] = 2[38.5 - \boxed{24.5}] = \boxed{28}$ cm$^2$

**Q6** **A $\boxed{\text{square}}$ sheet of side $28$ cm is folded into a cylinder by joining its two sides. Find the base area of the cylinder thus formed.**

**ANSWER / SOLUTION**

Side of square $= 28$ cm $\implies$ Circumference $2\pi r = 28$ cm

$2 \times \frac{22}{7} \times r = 28 \implies r = \frac{28 \times 7}{2 \times 22 \times \boxed{1}} = \boxed{\frac{49}{11}}$ cm

Base area of cylinder $= \pi r^2 = \frac{22}{7} \times \frac{49}{11} \times \frac{\boxed{49}}{\boxed{11}} = \boxed{\frac{686}{11}}$ cm$^2$

## Q.3 (B) Solve / Prove

3 Marks each

5 QUESTIONS · 1 ☐ Revised

**Q1** **In a clock, the minute hand is of length $7$ cm. Find the area covered by the minute hand in 5 minutes.**

**ANSWER / SOLUTION**

Length of minute hand $r = 7$ cm.

Angle rotated in 60 minutes $= 360^\circ \implies$ Angle in 5 minutes $\theta = \frac{5}{60} \times 360^\circ = 30^\circ$.

Area covered $=$ Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 7^2 = \frac{1}{12} \times 154 = 12.83$ cm$^2$

**Q2** **Three cubes each of side $15$ cm joined end to end. Find the total surface area of the resulting cuboid.**

**ANSWER / SOLUTION**

Length $l = 15 + 15 + 15 = 45$ cm, breadth $b = 15$ cm, height $h = 15$ cm.

Total surface area $= 2(lb + bh + hl) = 2(45 \times 15 + 15 \times 15 + 15 \times 45) = 2(675 + 225 + 675) = 2(1575) = 3150$ cm$^2$

**Q3** **The radii of the circular ends of a frustum of a cone are $14$ cm and $8$ cm. If the height of the frustum is $8$ cm find: (a) Slant height of frustum. (b) Total surface area of frustum. (c) Volume of frustum ($\pi = 3.14$).**

TEXTBOOK **Solved Example · Ex. (2) · Page 147**

**ANSWER / SOLUTION**

$r_1 = 14$ cm, $r_2 = 8$ cm, $h = 8$ cm.

**(a) Slant height $l$:** $l = \sqrt{h^2 + (r_1 - r_2)^2} = \sqrt{8^2 + (14 - 8)^2} = \sqrt{64 + 36} = 10$ cm

**(b) TSA of frustum:** $\pi l(r_1 + r_2) + \pi r_1^2 + \pi r_2^2 = 3.14[10(22) + 196 + 64] = 3.14[220 + 260] = 3.14 \times 480 = 1507.2$ cm$^2$

**(c) Volume of frustum:** $\frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2) = \frac{1}{3} \times 3.14 \times 8 \times (196 + 64 + 112) = \frac{25.12}{3} \times 372 = 3114.88$ cm$^3$

**Q4** A sector of a circle of radius $15$ cm has the angle $120^\circ$. It is rolled up so that two bounding radii are joined together to form a cone. Find the volume of the cone ($\pi = \frac{22}{7}$).

**ANSWER / SOLUTION**

Slant height of cone $l$ = radius of sector $= 15$ cm.

Length of arc of sector $= \frac{120^\circ}{360^\circ} \times 2\pi(15) = 10\pi$ cm.

Circumference of base of cone $2\pi r = 10\pi \implies r = 5$ cm.

Height of cone $h = \sqrt{l^2 - r^2} = \sqrt{15^2 - 5^2} = \sqrt{200} = 10\sqrt{2}$ cm $\approx 14.14$ cm.

Volume of cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 25 \times 14.14 = 370.37$ cm$^3$

**Q5** A metallic sphere of radius $10.5$ cm is melted and then recast into small cones, each of radius $3.5$ cm and height is $3$ cm. Find how many cones are obtained?

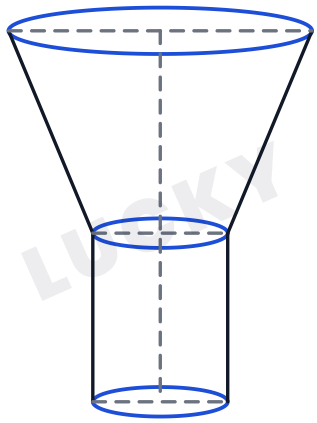
**ANSWER / SOLUTION**

Number of cones $= \frac{\text{Volume of sphere}}{\text{Volume of one cone}} = \frac{\frac{4}{3}\pi R^3}{\frac{1}{3}\pi r^2 h} = \frac{4\times(10.5)^3}{(3.5)^2\times 3} = \frac{4\times 1157.625}{12.25\times 3} = \frac{4630.5}{36.75} = 126$

$\therefore$ 126 cones are obtained.

## Q.4 Long Solve / Prove — 4 Marks each — 4 QUESTIONS — ☐ Revised

**Q1** An oil funnel made of tin sheet consists of a cylindrical portion $10$ cm long attached to a frustum of a cone. If the total height is $22$ cm, diameter of the cylindrical portion is $8$ cm and the diameter of the top of funnel is $18$ cm. Find the area of tin required to make the funnel.



**ANSWER / SOLUTION**

Cylinder: $r = 4$ cm, $h = 10$ cm. Frustum: $r_1 = 9$ cm, $r_2 = 4$ cm, $H = 22 - 10 = 12$ cm.

Slant height of frustum $l = \sqrt{H^2 + (r_1 - r_2)^2} = \sqrt{12^2 + (9-4)^2} = \sqrt{144 + 25} = 13$ cm.

Area of tin required = CSA of cylinder + CSA of frustum $= 2\pi rh + \pi(r_1 + r_2)l$

$= \pi[2(4)(10) + (9+4)(13)] = \pi[80 + 169] = \frac{22}{7} \times 249 = 782.57$ cm$^2$

**Q2** A toy is in the form of a cone mounted on a hemisphere. The diameter of the base of the cone and that of a hemisphere is $18$ cm and the height of cone is $12$ cm. Find the total surface area of toy ($\pi = 3.14$).

**ANSWER / SOLUTION**

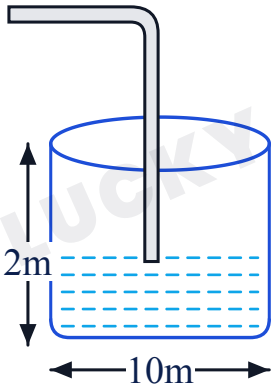
Radius $r = 9$ cm, height of cone $h = 12$ cm.

Slant height of cone $l = \sqrt{r^2 + h^2} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = 15$ cm.

TSA of toy = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2 = \pi r(l + 2r)$

$= 3.14 \times 9 \times (15 + 2 \times 9) = 28.26 \times 33 = 932.58$ cm$^2$

**Q3** A farmer connects a pipe of internal diameter $20\ \text{cm}$ from the canal into a cylindrical tank in his field, which is $10\ \text{m}$ in diameter and $2\ \text{m}$ deep. If water flows through the rate of $3\ \text{km/h}$, in how much time will the tank be filled?




**ANSWER / SOLUTION**

Pipe radius $r = 10\ \text{cm} = 0.1\ \text{m}$. Tank radius $R = 5\ \text{m}$, depth $H = 2\ \text{m}$.

Volume of tank $= \pi R^2 H = \pi \times 5^2 \times 2 = 50\pi\ \text{m}^3$.

Speed of water $= 3\ \text{km/h} = 3000\ \text{m/h}$. Volume flowing per hour $= \pi r^2 h = \pi \times (0.1)^2 \times 3000 = 30\pi\ \text{m}^3/\text{h}$.

Time required $= \frac{50\pi}{30\pi} = \frac{5}{3}$ hours $= 1$ hour 40 minutes $= 100$ minutes.

**Q4** How many coins $1.75\ \text{cm}$ in diameter and $2\ \text{mm}$ thick must be melted to form a cuboid $11\ \text{cm} \times 10\ \text{cm} \times 7\ \text{cm}$?

**ANSWER / SOLUTION**

Coin radius $r = \frac{1.75}{2} = 0.875\ \text{cm}$, thickness $h = 2\ \text{mm} = 0.2\ \text{cm}$.

Volume of one coin $= \pi r^2 h = \frac{22}{7} \times (0.875)^2 \times 0.2 = 0.48125\ \text{cm}^3$.

Volume of cuboid $= 11 \times 10 \times 7 = 770\ \text{cm}^3$.

Number of coins $= \frac{\text{Volume of cuboid}}{\text{Volume of one coin}} = \frac{770}{0.48125} = 1600$

## Q.5 Creative / HOTS

3 Marks each — 5 QUESTIONS — ☐ Revised

**Q1 (a)** **Write True or False and justify your answer in the following:** A solid ball is exactly fitted inside the cubical box of side $b$. The volume of ball is $\frac{4}{3}\pi b^3$.

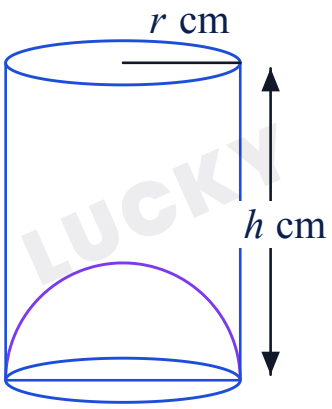
**ANSWER / SOLUTION**

**False**

**Justification:** If a sphere fits exactly inside a cubical box of side $b$, its diameter equals $b$, so radius $r = \frac{b}{2}$.

Volume of ball $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left(\frac{b}{2}\right)^3 = \frac{4}{3}\pi\frac{b^3}{8} = \frac{1}{6}\pi b^3 \neq \frac{4}{3}\pi b^3$.

**Q1 (b)** **Write True or False and justify your answer in the following:** The capacity of cylindrical vessel with a hemispherical portion raised upward at the bottom as shown in fig. is $\frac{1}{3}\pi r^2(3h - 2r)$ where $r$ is radius in $\text{cm}$ and $h\ \text{cm}$ is height.




**ANSWER / SOLUTION**

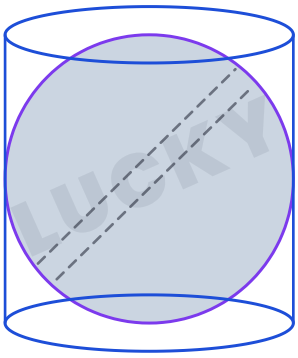
**True**

**Justification:** Capacity

$=$ Volume of cylinder $-$ Volume of hemisphere $= \pi r^2 h - \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2(3h - 2r)$.

**Q2** In the figure, a sphere is placed in a cylinder. It touches the top, bottom and curved surface of the cylinder. If the radius of the base of the cylinder is '$r$', answer the following:

a. What is the height of the cylinder in terms of '$r$'?
b. What is the ratio of the curved surface area of the cylinder and the surface area of the sphere?
c. What is the ratio of volumes of the cylinder and of the sphere?




**ANSWER / SOLUTION**

**(a) Height of cylinder:** $h = 2r$

**(b) Ratio of CSA:** $\frac{\text{CSA of cylinder}}{\text{Surface area of sphere}} = \frac{2\pi rh}{4\pi r^2} = \frac{2\pi r(2r)}{4\pi r^2} = \frac{4\pi r^2}{4\pi r^2} = 1:1$

**(c) Ratio of Volumes:** $\frac{\text{Volume of cylinder}}{\text{Volume of sphere}} = \frac{\pi r^2 h}{\frac{4}{3}\pi r^3} = \frac{\pi r^2(2r)}{\frac{4}{3}\pi r^3} = \frac{2\pi r^3}{\frac{4}{3}\pi r^3} = \frac{6}{4} = 3:2$

**Q3** A horse is tied to a peg at one corner of square shaped grass field of side $15\ \text{m}$ by means of $5\ \text{m}$ long rope, find:

a. The area of that part of the field in which the horse can graze.
b. The increase in the grazing area if the rope were $10\ \text{m}$ long instead of $5\ \text{m}$ ($\pi = 3.14$).

**ANSWER / SOLUTION**

**(a) Grazing area with $5\ \text{m}$ rope:** Sector angle $\theta = 90^\circ$.

$\text{Area} = \frac{90^\circ}{360^\circ} \times \pi r^2 = \frac{1}{4} \times 3.14 \times 5^2 = \frac{78.5}{4} = 19.625\ \text{m}^2$

**(b) Increase in grazing area with $10\ \text{m}$ rope:**

$\text{New Area} = \frac{1}{4} \times 3.14 \times 10^2 = \frac{314}{4} = 78.5\ \text{m}^2$

$\text{Increase in area} = 78.5 - 19.625 = 58.875\ \text{m}^2$

**Q4** A donor agency ensures milk is supplied in containers, which are in the form of a frustum of cone to be distributed to flood victims in a camp. The height of each frustum is $30\ \text{cm}$ and the radii of lower and upper circular ends are $20\ \text{cm}$ and $40\ \text{cm}$ respectively. If this milk is available at the rate of ₹ $35$ per litre and $8800$ litres of milk is needed daily for a camp:

a. Find how many milk containers are needed daily for the camp.
b. What daily cost will put on the donor agency?

**ANSWER / SOLUTION**

Volume of 1 frustum container $= \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2) = \frac{1}{3} \times \frac{22}{7} \times 30 \times (40^2 + 20^2 + 40 \times 20)$

$= \frac{220}{7} \times (1600 + 400 + 800) = \frac{220}{7} \times 2800 = 88000\ \text{cm}^3 = 88\ \text{litres}$

**(a) Number of containers needed:** $\frac{8800}{88} = 100\ \text{containers}$

**(b) Daily cost:** $8800\ \text{litres} \times ₹\ 35/\text{litre} = ₹\ 3,08,000$

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